

Outlier remove data cleaning

Exercise

(Normal Distribution and Z Score for Outlier Removal)

You are given bhp.csv which contains property prices in the city of banglore, India.

You need to examine price_per_sqft column and do following,

1. Remove outliers using percentile technique first. Use [0.001, 0.999] for lower and upper bound percentiles
2. After removing outliers in step 1, you get a new dataframe.
3. On step(2) dataframe, use 4 standard deviation to remove outliers
4. Plot histogram for new dataframe that is generated after step (3). Also plot bell curve on same histogram
5. On step(2) dataframe, use zscore of 4 to remove outliers. This is quite similar to step (3) and you will get exact same result

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Solution for data Cleaning

https://github.com/codebasics/math-for-machine-learning/blob/main/3_normal_distribution/Exercise/exercise_solution.ipynb

```
import pandas as pd
import matplotlib
from matplotlib import pyplot as plt
df = pd.read_csv("bhp.csv")
df.head()
```

We will quickly examine price_per_sqft column to check min,max,mean and percentile values

```
df.price_per_sqft.describe()
```

Above, you can see max is 12000000, which is too much price for price per sqft. Looks like this dataset has some very bad outliers so we will first remove those with percentile and then use z score

Also plot histogram to see how is the data distribution

```
plt.hist(df.price_per_sqft, bins=20, rwidth=0.8)
plt.xlabel('Price per square ft')
plt.ylabel('Count')
plt.show()
```

Above chart confirms our understanding that this dataset has some bad outliers.

Because we see only one bar with more than 12000 data points but on the right hand side there are data points that have higher value for price per sqft but count of these data points is very tiny compared to 12000 hence we can hardly see any bar

Using log scale can make visualization little better

```
plt.hist(df.price_per_sqft, bins=20, rwidth=0.8)
plt.xlabel('Price per square ft')
plt.ylabel('Count')
plt.yscale('log')
plt.show()
```

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Treat outliers using percentile first

```
lower_limit, upper_limit = df.price_per_sqft.quantile([0.001, 0.999])
lower_limit, upper_limit
```

```
outliers = df[(df.price_per_sqft>upper_limit) | (df.price_per_sqft<lower_limit)]
outliers.sample(10)
```

Above shows properties that has price_per_sqft value such as 500 rs which is very very low for the city of banglore. You would be very lucky if you can find such as property :)

Also we have properties that has price per sqft value such as 12000000 which is way too high.

It seems almost impossible and even if it is legitimate it can help data analysis process if we remove such a data point

```
df2 = df[(df.price_per_sqft<upper_limit) & (df.price_per_sqft>lower_limit)]
df2.shape
```

```
df.shape
```

```
df.shape[0] - df2.shape[0]
```

We removed total 28 outliers

Now remove outliers using 4 standard deviation

```
max_limit = df2.price_per_sqft.mean() + 4*df2.price_per_sqft.std()
min_limit = df2.price_per_sqft.mean() - 4*df2.price_per_sqft.std()
max_limit, min_limit
```

```
df2[(df2.price_per_sqft>max_limit) | (df2.price_per_sqft<min_limit)].sample(10)
```

```
df3 = df2[(df2.price_per_sqft>min_limit) & (df2.price_per_sqft<max_limit)]
df3.shape
```

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```
df2.shape[0]-df3.shape[0]
```

In this step we removed total 125 outliers

```
plt.hist(df3.price_per_sqft, bins=20, rwidth=0.8)
plt.xlabel('Price per square ft')
plt.ylabel('Count')
plt.show()
```

Now remove outliers using z score. Use z score of 4 as your threshold

```
df2['zscore'] = (df2.price_per_sqft-
df2.price_per_sqft.mean())/df2.price_per_sqft.std()
df2.sample(10)
```

```
outliers_z = df2[(df2.zscore < -4) | (df2.zscore>4)]
outliers_z.shape
```

```
outliers_z.sample(5)
```

```
df4 = df2[(df2.zscore>-4)&(df2.zscore<4)]
df4.shape
```

```
df2.shape[0] - df4.shape[0]
```